THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 07

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For the LCM of 15 and 25, Student A gets 75 and Student B gets 375. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

2. A student states: "A composite number can be factorised into primes in more than one way (ignoring order)." The correct response is: 1

(a) Correct

(b) Correct for even numbers

(c) Incorrect — the factorisation is unique (FTA)

(d) Cannot be determined

3. The zeroes of $x^2 + 4x - 21$ are: 1

(a) $-3, 7$ **(b)** $3, -7$ **(c)** $-21, 1$ **(d)** $7, 3$

4. If the sum of the zeroes of $kx^2 - 3x + 5$ is 1, then k is: 1

(a) $\frac{1}{3}$ **(b)** 3 **(c)** 5 **(d)** −3

5. The pair $5x - 8y + 1 = 0$ and $3x - \frac{24}{5}y + \frac{3}{5} = 0$ has: 1

(a) infinitely many solutions **(b)** two solutions

(c) a unique solution **(d)** no solution

6. A student states: "A pair of linear equations can have exactly two solutions." The correct response is: 1

(a) Correct

(b) Incorrect — it has 0, 1 or infinitely many solutions

(c) Correct for parallel lines

(d) Cannot be determined

7. Which of the following is **not** a quadratic equation? 1

(a) $x(x+1) = 6$ **(b)** $x^2 = 5x$ **(c)** $2x^2 + 1 = 0$ **(d)** $(x+2)^2 = x^2 + 3$

8. "The area of a square of side $(x + 3)$ cm is 49 cm²" is represented by: 1

(a) $x^2 + 3x - 49 = 0$ **(b)** $x^2 + 6x - 40 = 0$ **(c)** $x^2 - 40 = 0$ **(d)** $x^2 + 6x + 9 = 0$

9. The 20th term of the AP $4, 9, 14, \ldots$ is: 1

(a) 94 **(b)** 99 **(c)** 100 **(d)** 104

10. The sum of the first 20 odd natural numbers is: 1

(a) 420 **(b)** 200 **(c)** 380 **(d)** 400

11. If $\triangle ABC \sim \triangle DEF$ with $AB = 4$ cm and $DE = 6$ cm, then $\frac{BC}{EF}$ is: 1

(a) $\frac{4}{9}$ **(b)** $\frac{3}{2}$ **(c)** 1 **(d)** $\frac{2}{3}$

12. A student states: "All isosceles triangles are similar." The correct response is: 1

(a) Correct

(b) Correct only for right triangles

(c) Incorrect — their angles need not be equal

(d) Cannot be determined

13. The value of $\sin 45^\circ \cos 45^\circ$ is: 1

(a) $\frac{1}{2}$ **(b)** $\sqrt{2}$ **(c)** $\frac{1}{\sqrt{2}}$ **(d)** 1

14. The length of the tangent from a point 5 cm from the centre of a circle of radius 3 cm is: 1

(a) 4 cm **(b)** $\sqrt{34}$ cm **(c)** 2 cm **(d)** 8 cm

15. The length of an arc of radius 21 cm subtending 120° is: 1

(a) 132 cm **(b)** 44 cm **(c)** 66 cm **(d)** 22 cm

16. The perimeter of a semicircular protractor of diameter 14 cm is: 1

(a) 44 cm **(b)** 36 cm **(c)** 29 cm **(d)** 22 cm

17. The volume of a hemisphere of radius 3 cm is: 1

(a) 18π cm³ **(b)** 36π cm³ **(c)** 27π cm³ **(d)** 9π cm³

18. If the median is 15 and the mean is 16, the mode is: 1

(a) 13 **(b)** 17 **(c)** 18 **(d)** 14

19. The median of 3, 5, 7, 9, 11 is: 1

(a) 5 **(b)** 7 **(c)** 9 **(d)** 35

20. The probability of drawing a face card from a deck of 52 cards is: 1

(a) $\frac{4}{13}$ **(b)** $\frac{3}{13}$ **(c)** $\frac{1}{13}$ **(d)** $\frac{1}{2}$

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Find the HCF and LCM of 8, 9 and 25. 2

22. If one zero of $2x^2 + 3x + k$ is $\frac{1}{2}$, find k and the other zero. 2

23. For what value of p does $px^2 - 4x + 1 = 0$ have equal roots? 2

24. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2.4$ cm, $AE = 3.2$ cm and $EC = 4.8$ cm. Find DB. 2

25. Find the coordinates of the point dividing the join of A(1, 3) and B(7, 9) in the ratio 1 : 2. 2

26. If $5\tan\theta = 4$, find $\frac{5\sin\theta - 3\cos\theta}{5\sin\theta + 2\cos\theta}$. 2

27. A tower is $50\sqrt{3}$ m high. The angle of elevation of its top from a point on the ground is 60°. Find the distance of the point from the tower. 2

28. A point P is 17 cm from the centre of a circle of radius 8 cm. Find the length of the tangent PT and $\angle OTP$. 2

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SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $2x + y = 4$ and $x - y = -1$. 3

30. Find the median of: 3

Class	0–10	10–20	20–30	30–40	40–50
Frequency	8	12	10	6	4

31. How many three-digit numbers are divisible by 7? 3

32. Prove that in two concentric circles, all chords of the larger circle which touch the smaller circle are equal in length. 3

33. A sector of a circle of radius 15 cm has angle $120°$. Find its area, arc length and perimeter. (Use $\pi = 3.14$) 3

34. Cards numbered 1 to 25 are in a box. One card is drawn. Find P(odd number), P(multiple of 3) and P(prime). 3

35. As observed from the top of a 75 m high lighthouse, the angles of depression of two ships are $30°$ and $45°$. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the ships. 3

OR

Two poles of equal heights stand on either side of a road 80 m wide. At a point between them on the road, the angles of elevation of the tops of the poles are $60°$ and $30°$. Find the height of the poles and the distances of the point from the poles. 3

36. A student found the mean of the following data to be 52. Verify his answer. 3

Class	10–30	30–50	50–70	70–90	90–110	110–130
Frequency	5	8	12	20	3	2

OR

Heights of 51 girls are given in "less than" form. Find the median height. 3

Height less than (cm)	140	145	150	155	160	165
No. of girls	4	11	29	40	46	51

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Find the circumcentre and circumradius of the triangle with vertices $(8, 6)$, $(8, -2)$ and $(2, -2)$. 5

OR

(i) Find the ratio in which the y-axis divides the segment joining $(-2, -3)$ and $(3, 7)$, and the point of division. 5
(ii) Find the length of the segment.

38. Prove that $\frac{1}{\sec A - \tan A} - \frac{1}{\cos A} = \frac{1}{\cos A} - \frac{1}{\sec A + \tan A}$. 5

OR

Prove that $\left(1 + \frac{1}{\tan^2 A}\right)\left(1 + \frac{1}{\cot^2 A}\right) = \frac{1}{\sin^2 A - \sin^4 A}$. 5

39. A hemispherical depression of radius 3.5 cm is cut into the top face of a cuboid 10 cm × 10 cm × 5 cm. Find **5**

(i) the volume of the remaining block

(ii) its total surface area.

OR

A solid is a cone of radius 5 cm and height 12 cm mounted on a hemisphere of the same radius. Find **5**

(i) its volume

(ii) its total surface area. (Use $\pi = 3.14$)

40. Find the median and the mode of daily wages: **5**

Wages (₹)	100–150	150–200	200–250	250–300	300–350
Workers	6	10	15	12	7

OR

Case study: The life of 60 batteries (in hours) is given below. Find **5**

(i) the mean

(ii) the median

(iii) the mode.

Life (hrs)	50–60	60–70	70–80	80–90	90–100
Batteries	6	10	18	16	10

END OF QUESTION PAPER **Total: 80 Marks**

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